

Finite Configurations Cannot Generate a Constant Trace in Rule 30

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For every right half we determine the unique left half whose Rule-30 central trace is constant: an alternating tail selected by the leading one of the right half when the initial center is 0, and a single universal checkerboard when it is 1. Every nonzero member of either fiber carries infinitely many ones, so the zero row is the only finite configuration with a constant trace. For support radius w the sharp maximum constant-prefix length is $2\lceil w/2 \rceil + 1$ when the initial center is 0 and $2\lfloor w/2 \rfloor + 2$ when it is 1, so the maximum over both is $w + 2$, attained by exactly 2^w configurations for even w and $2^w - 1$ for odd w . Hence no column of a nonzero finite Rule 30 orbit is eventually constant.

Keywords. cellular automata; Rule 30; temporal trace; finite configuration; symbolic dynamics

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1 Introduction

Rule 30 is the binary cellular automaton on $\{0, 1\}^{\mathbb{Z}}$ with local rule

$$F(x)_i = x_{i-1} \oplus (x_i \vee x_{i+1}), \quad (1)$$

where $\oplus$ is addition modulo 2 and $\vee$ is disjunction. It is left permutive: fixing x_i and x_{i+1} , the value $F(x)_i$ is a bijection of x_{i-1} .

Wolfram's Prize Problem 1 asks whether the center trace of the lone seed is eventually periodic [8], restated as Problem 3.10 of [4]; Kopra records the broader finite-configuration version as open [5]. Jen proved that a finite Rule 30 orbit has at most one eventually periodic column [3, Thm. 2b][2, Prop. 3], and Kopra's width-two theorem does not decide whether that exceptional column exists [4]. The obstruction is structural. Rule 90 is left permutive as well, and it admits finite configurations whose center trace is identically zero (Figure 1(b)), so no argument resting on permutivity alone separates the two rules. The classification below turns on the OR of (1), which latches, where Rule 90's XOR does not.

We settle the eventual-period-one subcase for every nonzero finite configuration, a class strictly broader than the lone seed. For each right half exactly one left half gives a constant center trace: an alternating tail selected by the leading one of the right half when the center starts at 0, and one universal checkerboard when it starts at 1 (Theorems 2 and 3). Every nonzero member of either fiber carries infinitely many ones, so the zero row is the only finite configuration with a constant trace (Corollaries 4 and 5). A row of support radius w holds its center constant for at most $2\lceil w/2 \rceil + 1$ steps when the center starts at 0 and $2\lfloor w/2 \rfloor + 2$ when it starts at 1, hence $w + 2$ overall, attained by exactly 2^w rows for even w and $2^w - 1$ for odd w (Theorems 6 and 7, Corollary 8). This does not resolve Prize Problem 1, and every nonconstant eventual period remains open.

Demanding an all-zero center forces the left half cell by cell via exact inversion, acting as a prefix-OR of the prescribed right half. A right half whose *first* one sits at depth 3 forces $L_1, L_2, \dots = 0, 0, 1, 0, 1, 0, 1, \dots$, ones at every subsequent odd depth. This one-sided bijectivity mirrors Rowland's unique-history construction [6, Prop. 1]. The latching OR in (1) makes the classification Rule-30-specific, unlike Rule 90 (Figure 1, Section 3).

Jen gives necessary and sufficient conditions for a finite initial condition to generate an eventually-one temporal sequence [3, Thm. 7a]. Rule 30 meets the hypothesis $a_0 = 0, a_1 = a_4 = 1$ but none of the

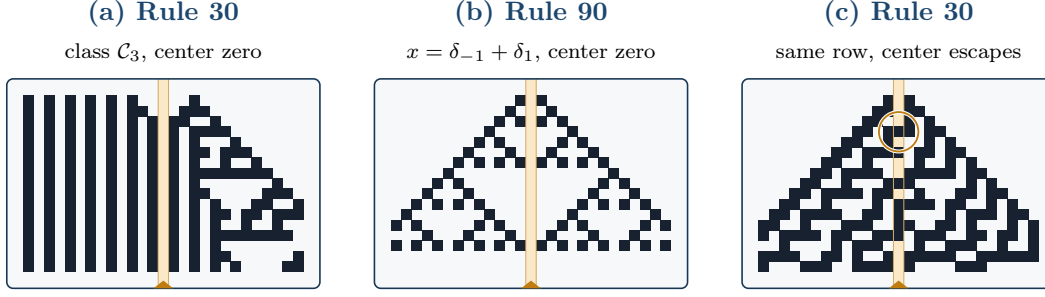


Figure 1: A zero center trace forces an infinite left tail under Rule 30, but not under Rule 90. Time runs downward; the gold band and caret mark column 0. (a) The left half forced by a right half of class $\mathcal{C}_3$ alternates, hence carries infinitely many ones, which is what Corollary 4 exploits. (b) and (c) run the single finite row $x = \delta_{-1} + \delta_1$ under Rule 90 and under Rule 30: its center stays zero under Rule 90 and escapes under Rule 30. The only difference between the two rules is the OR latch of (1).

four alternatives, which require respectively $a_6 = a_7 = 1$, $a_6 = 1$, $a_7 = 1$, or $a_5 = 1$, while Rule 30 has $a_5 = a_6 = a_7 = 0$; necessity therefore already excludes an eventually-one column for every finite Rule 30 configuration, and we claim no novelty for that exclusion. New here are the two exact fixed-locus fibers, the explicit zero exclusion, and the sharp horizons with extremizer counts. Jen remarks that symmetric results hold for constant-zero sequences but states neither a zero-fiber classification nor any of these quantitative conclusions. Finally, deciding whether a subconfiguration of length K has a K -step preimage is NP-complete already for a nearest-neighbor rule [1], and one-step inversion has been solved for the *additive* class over $\mathbb{Z}_p$ by an operator analogous to backward integration [7]; Theorem 2 inverts a constraint imposed over all future time, for a rule that is not additive.

2 The constant-trace fibers

The trace and the two halves of a row are written as follows.

$$c_t := \text{Tr}(x)_t = F^t(x)_0, \quad R_j := x_j, \quad L_j := x_{-j} \quad (j \geq 0) \quad (2)$$

Throughout, $\mathcal{F}$ denotes the finite-support configurations.

The trace sees only column 0, but under a left-permutive rule a discrepancy left of the origin propagates rightward along a diagonal and must eventually reach it. The classification rests on this one fact, which uses only locality and left permutivity, not the specific form of (1).

Lemma 1 (Triangular uniqueness). *If two configurations agree at every coordinate $i \geq 0$ and have the same central trace, then they are equal.*

Proof. If $-n$ is their rightmost differing coordinate, locality preserves agreement to its right through time n . Left permutivity preserves the discrepancy along $(-n, 0), (-n+1, 1), \dots, (0, n)$, contradicting equal traces. In its finite form, if the rows agree on $i \geq 0$ and their traces agree for $0 \leq t \leq r$, the same induction forces agreement at $-r, -r+1, \dots, -1$.

Uniqueness alone does not say what the forced left half looks like. The next result computes it: the rows with zero trace form one invariant class $\mathcal{C}_m$ for each position m of the first right-hand one, each class is carried to its predecessor by F , and the left half is a prefix-OR of the right. Figure 1(a) is a member of $\mathcal{C}_3$.

Theorem 2 (Zero-trace fiber). *For a prescribed right half with $R_0 = 0$, the zero-trace-compatible left half is unique. Suppose the positive half is nonzero, and let $m = \min\{j \geq 1 : R_j = 1\}$. The forced left half is then the following, with R_j for $j > m$ arbitrary.*

$$L_j = 0 \ (j < m), \quad L_m = 1, \quad L_j = j \bmod 2 \ (j > m) \quad (3)$$

If instead the positive half is zero, the unique row is zero. The same classification in prefix-OR form is the following.

$$L_{2k+1} = \bigvee_{j=1}^{2k+1} R_j \quad (k \geq 0) \quad (4)$$

$$L_{2k} = R_{2k} \wedge \neg \bigvee_{j=1}^{2k-1} R_j \quad (k \geq 1) \quad (5)$$

Proof. For $m \geq 1$, let $\mathcal{C}_m$ consist of rows satisfying these conditions. One Rule 30 step gives the three identities below.

$$\begin{aligned} x'_0 &= L_1 \oplus R_1 \\ R'_j &= R_{j-1} \oplus (R_j \vee R_{j+1}) \\ L'_j &= L_{j+1} \oplus (L_j \vee L_{j-1}) \end{aligned} \quad (6)$$

Suppose $m > 1$. These identities then give the following.

$$x'_0 = 0, \quad R'_j = L'_j = 0 \ (j < m-1), \quad R'_{m-1} = L'_{m-1} = 1, \quad L'_m = m \bmod 2 \quad (7)$$

For $j > m$, including $j = m+1$ because $L_m = 1$, the OR in the last update is one, so $L'_j = (1 - (j \bmod 2)) \oplus 1 = j \bmod 2$. Thus $F(\mathcal{C}_m) \subseteq \mathcal{C}_{m-1}$. For $m = 1$, the alternating left half is fixed, $x'_0 = 1 \oplus 1 = 0$, and $R'_1 = 0 \oplus (1 \vee R_2) = 1$ independently of the right tail; hence $F(\mathcal{C}_1) \subseteq \mathcal{C}_1$. Every $\mathcal{C}_m$ therefore realizes the zero trace.

A nonzero right half selects a realization by its first one, and Lemma 1 makes it unique; a zero right half forces a zero left half by the same lemma against the zero row.

The other constant trace has a simpler fiber: one row per right half, and the left half no longer depends on which right half it is.

Theorem 3 (All-one fiber). *For a prescribed right half with $R_0 = 1$, the left half compatible with an identically one trace is unique and is the same for every such right half.*

$$L_j = 1 \iff j \text{ is positive and even} \quad (8)$$

Proof. Write $L_0 := x_0 = R_0 = 1$, so that the third identity of (6), $L'_j = L_{j+1} \oplus (L_j \vee L_{j-1})$, applies at every $j \geq 1$. Under (8) it gives $0 \oplus (1 \vee 0) = 1$ for even j and $1 \oplus (0 \vee 1) = 0$ for odd j , so the left half is fixed by F whatever the right half is. At the origin, $F(x)_0 = L_1 \oplus (L_0 \vee R_1) = 0 \oplus 1 = 1$ independently of R_1 , so the trace is identically one and $L_0 = 1$ is restored at every step. Lemma 1 makes the left half unique.

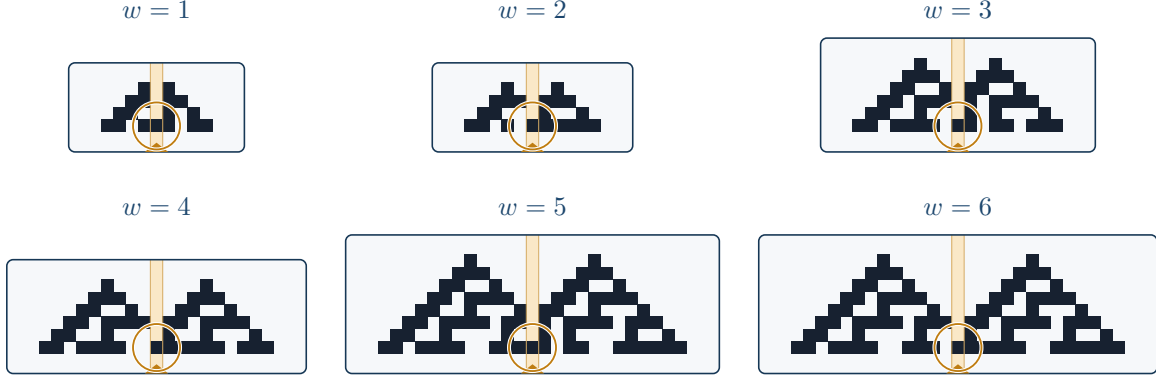


Figure 2: The zero-center horizon is flat across each pair $w = 2k - 1, 2k$, the parity effect of Theorem 6: raising the radius from odd to even buys no extra time. One extremizer per radius $w = 1, \dots, 6$, truncated from the zero-trace completion at left depth w . Time runs downward, and the circled cell is the first nonzero center, occurring at time $2\lceil w/2 \rceil + 1$.

3 Finite configurations

Every nonzero row in Theorem 2 carries ones arbitrarily far to the left, so none of them is finitely supported. That is the whole of the next statement, once one checks that a finite orbit cannot become the zero row.

Corollary 4. $\text{Tr}^{-1}(0^{\mathbb{N}}) \cap \mathcal{F} = \{0\}$. Moreover, no column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually zero.

Proof. Every nonzero row in Theorem 2 has ones at all sufficiently large odd left depths. If a column of a finite orbit were zero from time T onward, translation would make $F^T(x)$ a finite zero-trace row. It is nonzero: if $[a, b]$ is the support interval of a nonzero finite row, its next row has ones at $a - 1$ and $b + 1$ because Rule 30 maps 001 and 100 to one.

Corollary 4 is not a consequence of left permutivity alone: Rule 90 is also left permutive, yet the finite row with ones at ± 1 keeps a zero center forever (Figure 1(b,c)). The Rule-30-specific step is the OR latch $R'_1 = 1 \vee R_2 = 1$ in the proof of Theorem 2, which has no analogue in an additive rule.

The same latch upgrades the corollary from one constant to both, with no further machinery. Theorem 3 proves the second half again and independently, since (8) has ones at every positive even depth; the argument below is shorter and is the one to quote.

Corollary 5. No column of the orbit of a nonzero $x \in \mathcal{F}$ is eventually constant.

Proof. The eventually-zero case is Corollary 4. Suppose instead some column is eventually one; translating, take it to be column 0, so $c_t = 1$ for all $t \geq T$. Applying (1) at the origin gives $l_t = c_{t+1} \oplus (c_t \vee r_t)$, and $c_t = 1$ collapses the disjunction, so $l_t = 1 \oplus c_{t+1}$. For $t \geq T$ we also have $c_{t+1} = 1$, hence $l_t = 0$. Column -1 is then zero from time T onward, which Corollary 4 forbids.

4 A sharp finite horizon

Corollary 4 says a finite row's center cannot stay zero forever. How long it can stay zero, as a function of the support radius, is exactly one step past the light cone of the first odd depth beyond that radius.

For $w \geq 1$, let $\mathcal{S}_w = \{x \neq 0 : \text{supp}(x) \subseteq [-w, w], x_0 = 0\}$. For $x \in \mathcal{S}_w$, the zero-center horizon is defined as follows.

$$H(x) = \max\{h : F^t(x)_0 = 0 \text{ for } 0 \leq t \leq h\} \quad (9)$$

Thus $H(x)$ is the last matching time index; the corresponding prefix contains $H(x) + 1$ symbols, including the symbol at time zero.

Theorem 6 (Sharp finite horizon). *Let $w \geq 1$. Then both of the following hold.*

$$\max_{x \in \mathcal{S}_w} H(x) = 2 \left\lceil \frac{w}{2} \right\rceil, \quad \#\left\{x \in \mathcal{S}_w : H(x) = 2 \left\lceil \frac{w}{2} \right\rceil\right\} = 2^w - 1 \quad (10)$$

Proof. Let q be the least odd integer greater than w , so $q - 1 = 2\lceil w/2 \rceil$. If the positive right half is nonzero, a trace zero through time q , compared with its zero-trace completion, would force by finite-prefix uniqueness and (4) $x_{-q} = \bigvee_{j=1}^q R_j = 1$, contradicting $q > w$. If that half is zero, comparison with the zero row makes every nonzero left cell visible by time w . Thus $H(x) \leq q - 1$.

Conversely, choose any of the $2^w - 1$ nonzero words $(R_1, \dots, R_w)$, install (4)–(5) through left depth w , and zero all other cells. This row agrees with its infinite zero-trace completion through the light cone of time $q - 1$; when w is odd, the extra depth $w + 1$ is even and forced to zero. The missing odd cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

The one-trace horizon is the same computation against (8), and the parities exchange roles: ones sit at even depths, so the binding depth is the least *even* one past the radius, and $R_0 = 1$ already makes the row nonzero, which is why no extremizer is lost here.

Theorem 7 (Sharp finite horizon, one). *Let $w \geq 1$, let $\mathcal{T}_w = \{x : \text{supp}(x) \subseteq [-w, w], x_0 = 1\}$, and for $x \in \mathcal{T}_w$ let $H_1(x) = \max\{h : F^t(x)_0 = 1 \text{ for } 0 \leq t \leq h\}$. Let q be the least even integer greater than w . Then both of the following hold.*

$$\max_{x \in \mathcal{T}_w} H_1(x) = q - 1, \quad \#\{x \in \mathcal{T}_w : H_1(x) = q - 1\} = 2^w \quad (11)$$

Proof. A trace one through time q , compared with the all-one completion of Theorem 3, would force by finite-prefix uniqueness and (8) $x_{-q} = 1$, contradicting $q > w$; thus $H_1(x) \leq q - 1$. Conversely, choose any of the 2^w words $(R_1, \dots, R_w)$, install (8) through left depth w , and zero all other cells. This row agrees with its all-one completion through the light cone of time $q - 1$; when w is even, the extra depth $w + 1$ is odd and forced to zero. The missing even cell at depth q creates a discrepancy at time q , and finite-prefix uniqueness excludes further extremizers.

Combining the two horizons, the sharp law over both constants is flat in the parity that splits each one separately.

Corollary 8. *Let $w \geq 1$ and let $x \neq 0$ have $\text{supp}(x) \subseteq [-w, w]$. The longest constant prefix of the central trace of x has length at most $w + 2$, and exactly 2^w configurations attain length $w + 2$ when w is even, while exactly $2^w - 1$ attain it when w is odd.*

Proof. For $x_0 = 0$ Theorem 6 gives $2\lceil w/2 \rceil$, which is w for even w and $w + 1$ for odd w ; for $x_0 = 1$ Theorem 7 gives $q - 1$, which is $w + 1$ for even w and w for odd w . Hence the maximum horizon index is $w + 1$, so the maximum prefix length is $w + 2$. It is attained in the one case when w is even and in the zero case when w is odd, with the corresponding counts 2^w and $2^w - 1$.

5 Conclusion

Theorems 2 and 3 classify the two constant fibers of the trace map, and Corollaries 4 and 5 show that neither meets the finite configurations away from the origin. The next unresolved case is eventual period two; we make no claim about it or any higher period.

One structural fact bounds what this method reaches. Write $H(p, w)$ for the sharp maximum length of an eventually p -periodic central prefix over rows of support radius w , so that Theorems 6 and 7 give $H(1, w) = w + 2$. At $p = 2$ the constant-case mechanism has no analogue: left permutivity supplies, for any prescribed trace of length N and any compatible right prefix, a unique left half and hence a finite row of support radius $\approx N$ realizing that trace through time N , so $H(2, w) \geq w$ for every w . This argument does not decide whether $H(2, w)$ stays within a constant of w . Exhaustion over all rows of radius $w \leq 10$ (Appendix A) gives $H(2, w) = 7, 7, 7, 7, 9, 10, 10, 15, 17, 17$, so $H(2, w) - w = 6, 5, 4, 3, 4, 4, 3, 7, 8, 7$: an irregular staircase, not the exact law $w + 2$ of the constant case. The $p = 1$ horizon is finite and sharp in w ; for $p = 2$ only linear growth from below is proved, and a structural account of period-two exclusion remains open.

A Reproducibility and formalization

The theorems above are proved analytically. The computations recorded here corroborate them and generate the figures; no proof depends on them.

An independent numerical audit reproduces the sharp horizon and the extremizer counts of Theorems 6 and 7. For support radius w , write H_0 and H_1 for the maximum horizon index (9) when the initial center is zero and one respectively, each with the number of configurations attaining it.

w	H_0	count	H_1	count	$\max H$	count
1	2	1	1	2	2	1
2	2	3	3	4	3	4
3	4	7	3	8	4	7
4	4	15	5	16	5	16
5	6	31	5	32	6	31
6	6	63	7	64	7	64
7	8	127	7	128	8	127

Prefix lengths are these indices plus one, so $\max H + 1 = w + 2$ at every radius. The two horizons alternate which parity leads: H_0 is larger for odd w , H_1 for even w . The final column confirms Corollary 8, 2^w extremizers for even w and $2^w - 1$ for odd w .

For period two, `period_two_horizon.py` enumerates all 2^{2w+1} rows of radius $w \leq 10$ (the zero row included, both phases of 0101... counted) and reports the longest prefix of the center column that alternates. The lengths, that is $H(2, w)$ in the convention of the Conclusion, are 7, 7, 7, 7, 9, 10, 10, 15, 17, 17 for $w = 1, \dots, 10$, attained by 1, 1, 1, 2, 7, 11, 67, 3, 1, 2 rows. These are computations, not a bound: they show that the period-two horizon sits above w by 3 to 8 on this range with no regular pattern, and they prove nothing for larger w .

Every cell in Figures 1 and 2 is generated from (1). Figure 1 exhibits an all-zero center for the displayed $\mathcal{C}_3$ row, an all-zero Rule-90 center for the row with ones at $\{-1, 1\}$, and escape of that same row under Rule 30. The six panels of Figure 2, for $w = 1, \dots, 6$, have first nonzero center times 3, 3, 5, 5, 7, 7.

The accompanying `Rule30ZeroTail.lean` is a partial formalization. It formalizes the local equation and left permutivity, propagation of a rightmost discrepancy and triangular unit sensitivity, the prefix-OR closed form assuming a first positive right-hand one, the local Boolean identities for the $\mathcal{C}_m$ invariant, and the non-left-bounded conclusion conditional on the classification disjunction. It does *not* derive that disjunction from an all-zero trace, and it does not formalize the all-one fiber, the finite-support corollaries,

or either sharp horizon and extremizer count. The formalization is therefore supporting verification, not a machine-checked proof of the main theorems.

Code availability. The audit script, the figure generator, the executable tests and the partial Lean formalization accompany this submission as ancillary files.

Use of AI tools. The model `anthropic/claude-opus-5` assisted with drafting, with the audit and figure scripts, and with locating references. The author derived and checked every definition, theorem, proof, and numerical value independently of these tools, verified every reference against its primary source, and takes full responsibility for all contents of this paper irrespective of how they were generated.

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